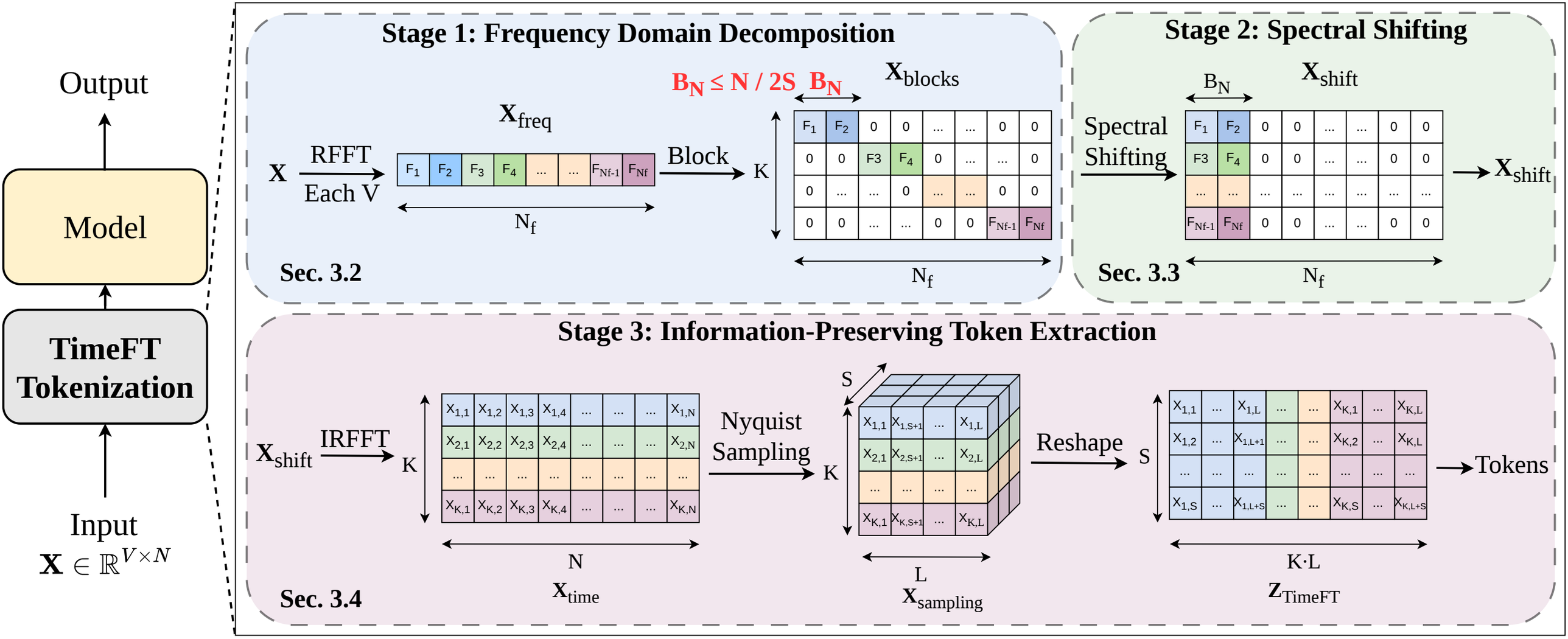

Output
Model
TimeFT Tokenization
Input
$\mathbf{X} \in \mathbb{R}^{V \times N}$
Stage 1: Frequency Domain Decomposition
$B_N \leq N / 2S$
$B_N$
$\mathbf{X}_{\text{blocks}}$
$\mathbf{X}_{\text{freq}}$
X
RFFT
Each V
Block
K
$N_f$
Sec. 3.2
Stage 2: Spectral Shifting
$B_N$
$\mathbf{X}_{\text{shift}}$
Spectral Shifting
$\mathbf{X}_{\text{shift}}$
$N_f$
Sec. 3.3
Stage 3: Information-Preserving Token Extraction
$\mathbf{X}_{\text{shift}}$
IRFFT
K
N
$\mathbf{X}_{\text{time}}$
Nyquist Sampling
S
K
L
$\mathbf{X}_{\text{sampling}}$
Reshape
S
K·L
$\mathbf{Z}_{\text{TimeFT}}$
Tokens
Sec. 3.4